

Urinalysis

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DEFINITION

Urinalysis is critical for evaluation of kidney and urinary tract disease. When a patient is first seen by a nephrologist, urinalysis must always be performed. Reagent strips are still the most widely used method for urinalysis to supply physicochemical information, but the nephrologist should be aware of their limitations and order more sensitive and specific measurements by other methods in the case of reagent strip abnormalities (e.g., the accurate measurement of proteinuria in case of reagent strip positivity for albumin).¹ Urine sediment examination is an integral part of urinalysis.^{2,3} For kidney patients, urine sediment examination should ideally be performed by nephrologists, who may be able to attribute clinical relevance to findings that may escape laboratory personnel.⁴

URINE SAMPLE

The use of an early morning urine sample is suggested by the Kidney Disease: Improving Global Outcomes (KDIGO) guideline,¹ especially for the measurement of albumin (see later).

However, the 24-hour urine collection is still widely used for the measurement of multiple parameters, even though errors caused by improper timing and missed samples can lead to overcollection or undercollection of urine. Errors can be minimized by giving the patient clear written instructions (e.g., at 7:00 AM discard the first urine of the morning, then collect in a capacious—of at least 2.5 L—and graduated container *all* the urine produced, including that passed at 7:00 AM of the day after; measure the volume of urine carefully and record it). Written instructions should also be provided for other types of urine samples (e.g., when testing for orthostatic proteinuria, one sample produced while the patient has been recumbent for some hours and another sample produced while the patient has been standing).

Strenuous physical exercise (e.g., running, soccer) should be avoided for at least 24 hours before the urine sample delivery to avoid exercise-induced proteinuria and hematuria or urinary casts. In women, urinalysis must be avoided during menstruation because of the high probability of blood contamination.

For urine microscopy, a midstream sample of the first morning urine is recommended by some international guidelines because this urine is the most concentrated and acidic and theoretically the best for the preservation of particles.⁵ On the other hand, the prolonged persistence in the bladder may favor the lysis of cells and casts, which may lead to false-negative results. For this reason, we use a combined dipstick and urine microscopy on the second morning urine.⁶

After handwashing, the external genitalia are washed and wiped dry with a paper towel. Then, women should spread the labia of the vagina and men withdraw the foreskin of the glans and collect the midstream urine after the first portion is discarded. The same procedures can be

used for children. For small infants, bags for urine are often used, even though these carry a high probability of contamination. A suprapubic bladder puncture may occasionally be necessary. In special situations, urine can also be collected through a bladder catheter, although this procedure may cause hematuria. Permanent indwelling catheters are almost invariably associated with bacteriuria, leukocyturia, hematuria, and candiduria.

The container for urine should be clean and have a capacity of at least 50 mL. It should have a wide base to avoid accidental spillage and should be capped. The label should identify the patient and the hour of urine collection.⁵

Several elements (but especially leukocytes) can lyse rapidly after collection; thus ideally the sample should be handled and examined as soon as possible. We recommend analysis within 3 hours from collection. If this is not possible, refrigeration of specimens at 4°C to 8°C assists preservation but may cause precipitation of phosphates or urates, which can hamper examination. Alternatively, chemical preservatives such as formaldehyde or glutaraldehyde can be used but can modify the urine pH, with consequent changes in urinary findings, and so cannot be universally recommended.

PHYSICAL CHARACTERISTICS

Color

The color of normal urine ranges from pale to dark yellow and amber, depending on the concentration of the urochrome. Abnormal changes in color can be caused by pathologic conditions, drugs, or foods (Table 4.1).

The most frequent pathologic conditions that can cause color changes of the urine are gross hematuria, hemoglobinuria, or myoglobinuria (pink, red, brown, or black urine); bilirubinuria (dark yellow to brown urine); and massive uric acid crystalluria (pink urine). Less frequent causes are urinary infection, mainly from *Klebsiella* spp., *Proteus mirabilis*, *Escherichia coli*, *Providencia stuartii*, or *Enterococcus* spp. in patients with permanent bladder catheter (purple urine, nowadays known as “purple urine bag syndrome”), chyluria (white milky urine), porphyrinuria (associated with the excretion in the urine of porphobilinogen), and alkaptonuria (red urine turning black over time).

The main drugs responsible for abnormal urine color are rifampin (yellow-orange to red urine); desferrioxamine (pinkish urine); phenytoin (red urine); chloroquine and nitrofurantoin (brown urine); triamterene, propofol, and blue dyes of enteral feeds (green urine); methylene blue (blue urine); and metronidazole, methyl dopa, and imipenem-cilastatin (urine darkening on standing).

Among the foods responsible are beetroot (red urine), senna and rhubarb (yellow to brown or red urine), and carotene (brown urine).

TABLE 4.1 Abnormal Changes in Urine Color and its Causes

Urine Color	Causes
Pink	Conditions: massive uric acid crystalluria, hematuria, hemoglobinuria, myoglobinuria Drugs: desferrioxamine
Purple	Conditions: urinary tract infection
Blue	Drugs: methylene blue
Green	Drugs: triamterene, propofol, blue dyes of enteral feeds
Orange	Conditions: bilirubinuria Drugs: rifampin
Red	Conditions: hematuria, hemoglobinuria, myoglobinuria, porphyrinuria Drugs: rifampin, phenytoin, phenazopyridine Foods: beetroot, senna, rhubarb
Brown	Conditions: hematuria, hemoglobinuria, myoglobinuria, bilirubinuria Drugs: chloroquine, nitrofurantoin Foods: carotene, senna, rhubarb
Black	Conditions: hematuria, hemoglobinuria, myoglobinuria, porphyrinuria, and alkaptonuria (red urine turning black over time) Drugs: metronidazole, methyl dopa, imipenem-cilastatin (urine darkens over time)
White	Conditions: chyluria

Turbidity

Normal urine is transparent. Urine can be turbid because of a high concentration of any urine particle, especially cells, crystals, and bacteria. The most frequent causes of turbidity are urinary tract infection (UTI), heavy hematuria, and genital secretions. Importantly, pathologic urine can be transparent.

Odor

A change in urine odor may be caused by the ingestion of some foods, such as asparagus. A pungent odor, caused by the production of ammonia, is typical of most bacterial UTIs, whereas there is often a sweet or fruity odor with ketones in the urine. Some rare conditions confer a characteristic odor to the urine. These include maple syrup urine disease (maple syrup odor), phenylketonuria (musty odor), isovaleric acidemia (sweaty feet odor), and hypermethioninemia (rancid butter or fishy odor).

Relative Density

Relative density can be measured by specific gravity or osmolality. *Specific gravity* (SG) refers to the weight of a volume of urine compared with the weight of the same volume of distilled water and depends on the mass and number of the dissolved particles. SG is most frequently evaluated by reagent strip (see discussion of chemical characteristics), which measures the ionic concentration of urine; alternatively, SG can be measured by refractometry. With reagent strip underestimation occurs with urine of pH above 6.5, whereas overestimation is found with urine protein concentration above 7.0 g/L.

For SG, we suggest refractometry for everyday practice because refractometers are inexpensive, simple to use, require only one drop of urine, and are more accurate than reagent strips. SG of 1.000 to 1.003

is seen with marked urinary dilution, as observed in patients with diabetes insipidus or water intoxication. SG of 1.010 is often called *isosthenuric* urine because it is of similar SG (and osmolality) to plasma, so it is often observed in conditions in which urinary concentration is impaired, such as acute tubular necrosis (ATN) and chronic kidney disease (CKD) with reduced glomerular filtration rate (GFR). SG above 1.040 almost always indicates the presence of some extrinsic osmotic agent, such as radiocontrast.

Osmolality is measured by an osmometer and depends only on the number of particles present and is not influenced by urine temperature or protein concentrations. Measurement of osmolality is hence more reliable than SG by either reagent strip or refractometry, although it can be influenced by high glucose concentrations.

CHEMICAL CHARACTERISTICS

Chemical characteristics of urine are most frequently evaluated by reagent strips. These plastic strips bear several pads (the most used are SG, pH, glucose, hemoglobin, albumin, leukocyte esterase, nitrites, bile pigments, and ketones), each pad being impregnated with chemical reagents meant to detect a specific urine feature. In clinical laboratories of wealthier countries, the reagent strip reading is performed by automated reader devices, using reflectance spectrometry, which supply highly reproducible results. Alternatively, the reading is performed manually, which is simple and quick but is exposed to subjectivity and incorrect procedures. Correct procedures imply rapid plunging of the strip in the urine; removal of the urine in excess on the pads to avoid color carryover from one pad to the close ones; adherence to the time interval between removal of the strip from urine and the reading of results as indicated by the manufacturer; and matching of the color developed in the pad with the color scale reported on the strip box under adequate light conditions.

Reagent strips have the advantages of simplicity and low cost and supply a full urinary profile within 2 to 3 minutes. Disadvantages include semiquantitative results only, susceptibility to interference by substances and urine discoloration. Sensitivity and specificity of reagent strips greatly differ across studies, which partly depend on the brand used (there is no standardization across manufacturers). [Table 4.2](#) summarizes the main causes of false-negative and false-positive results that can occur with strip reagent testing.

pH

The pH is determined by a strip that covers the pH range of 5.0 to 8.5 or to 9.0, with intervals of only 0.5, which limits precision. Moreover, significant deviations from true pH are observed for values below 5.5 and above 7.5. In the presence of formaldehyde, the strip supplies reduced pH values. When an accurate measurement of pH is necessary, a pH meter with a glass electrode is mandatory.

Urine pH reflects the presence of hydrogen ions (H^+), but this does not necessarily reflect the overall acid load in the urine because most of the acid is excreted as ammonia. Low pH is often observed with metabolic acidosis (in which acid is secreted), high-protein meals (which generate more acid and ammonia), and volume depletion (in which aldosterone is stimulated, resulting in an acid urine). High pH is often observed with renal tubular acidosis, vegetarian diets (caused by minimal nitrogen and acid generation), and infection because of urease-positive organisms that generate ammonia from urea.

Measurement of urine pH is also needed for a correct interpretation of other urine parameters (e.g., specific gravity, albumin) and several urine sediment findings, particularly for the evaluation of crystalluria.

TABLE 4.2 Urine Reagent Strip Testing

Constituent	False-Negative Results	False-Positive Results
Specific gravity (SG)	Urine pH > 6.5	Urine protein > 7.0 g/L
pH	Reduced values in presence of formaldehyde	—
Hemoglobin	High urine specific gravity Ascorbic acid Formaldehyde (0.5 g/L) used to preserve samples	Myoglobin Microbial peroxidases
Glucose	Ascorbic acid Bacteria	Very acidic urine pH Oxidizing detergents
Albumin	Low urine specific gravity Albumin < 0.25–0.30 g/L Low urine Tubular proteins Monoclonal heavy/light chains	Urine SG $\geq$ 1.030 Urine pH > 8.0 Quaternary ammonium detergents Chlorhexidine Polyvinylpyrrolidone
Leukocyte esterase	High urine specific gravity Ascorbic acid Glucose $\geq$ 20 g/L Protein > 5.0 g/L Cephalothin (strong inhibition) Tetracycline (strong inhibition) Cephalexin (moderate inhibition) Tobramycin (mild inhibition)	Formaldehyde (0.4 g/L) Imipenem Meropenem Clavulanate Abnormally colored urine
Nitrites	Bacteria that do not reduce nitrates to nitrites Lack of vegetables in diet Short bladder incubation time	Abnormally colored urine
Ketones	Improper storage	Free sulfhydryl groups (e.g., captopril) Levodopa Abnormally colored urine

Main causes of false-negative and false-positive results of urine reagent strips. False results also may occur when time-expired strips are used.

Hemoglobin

Hemoglobin is detected by a dipstick based on the pseudoperoxidase activity of the heme moiety of hemoglobin, which catalyzes the reaction of a peroxide and a chromogen to form a colored product.

False-negative results are most frequently caused by high SG or by ascorbic acid, a strong reducing agent, which can result in low-grade microhematuria being completely missed. Some reagent strips also include a vitamin C pad to reduce these false-negative results.⁷

The most important causes of false-positive results are myoglobinuria, resulting from rhabdomyolysis, and a high concentration of bacteria with pseudoperoxidase activity (*Enterobacteriaceae*, *Staphylococcus* spp., and *Streptococcus* spp.).⁸

Glucose

By reagent strip, with glucose oxidase as catalyst, glucose is first oxidized to gluconic acid and hydrogen peroxide. Through the catalyzing activity of a peroxidase, hydrogen peroxide then reacts with a reduced colorless chromogen to form a colored product. The reagent strip detects glucose concentrations of 0.5 to 20 g/L. When more precise quantification of urine glucose is needed, enzymatic methods such as hexokinase must be used. False-negative results for glucose occur in the presence of ascorbic acid and bacteria. False-positive findings may be observed in the presence of oxidizing detergents and very acidic urine pH.

Protein

Although historically the definition of proteinuria has varied, the definitions in the KDIGO guideline are increasingly used.¹ It is accepted

that physiologic proteinuria does not exceed 150 mg/24 hours for adults¹ and 140 mg/m² for children,⁹ in whom, however, the normal values do vary by age.¹ Three different approaches can be used for the evaluation of proteinuria, as described next.

Albumin Reagent Strip

The albumin reagent strip test is based on the effect of albumin on a buffer (tetrabromophenol blue), which causes a change in pH proportional to the concentration of the albumin itself. The strip is sensitive to albumin but has a very low sensitivity to other proteins, such as tubular proteins and light-chain immunoglobulins; thus it will not detect tubular proteinuria or overflow proteinuria, which can occur in monoclonal gammopathies. Moreover, the detection limit is 0.25 to 0.3 g/L, so it does not identify microalbuminuria and is influenced by hydration status (false-negative results may occur at low urine SG, and vice versa) and urine pH (false-positive results at strongly alkaline pH). The reagent strip supplies only a semiquantitative measurement of urine albumin, which is expressed on a scale from 0 to +++ or ++++. Some manufacturers also supply numerical results, although these represent only approximate quantitative measurements. Some reagent strips also include a creatinine pad, which supplies an albumin/creatinine ratio (ACR) and reduces the variability caused by changing diuresis and urine dilution.¹⁰ Nevertheless, for accurate quantification, other methods are needed.

24-Hour Protein Excretion

The 24-hour protein excretion averages the variation of proteinuria caused by the circadian rhythm and is still considered the reference

method.¹ The measurement of proteinuria can be done by chemical assays, turbidimetric techniques, or dye-binding techniques, which quantify total proteins rather than only albumin. However, the 24-hour urine collection can be impractical in some settings (e.g., children, outpatients, elderly patients) and is subject to error from overcollection or undercollection.¹¹

Protein/Creatinine Ratio and Albumin/Creatinine Ratio on Random Urine Sample

The protein/creatinine ratio (PCR) measured on an early morning urine sample represents a practical alternative to the 24-hour urine collection because the sample is easy to supply and is not influenced by variation in water intake or rate of diuresis.¹ The PCR is obtained by the ratio between urine protein excretion and creatinine excretion, expressed as milligrams per milligrams or milligrams per millimole. A close correlation between the PCR in a random urine sample and the 24-hour protein excretion has been demonstrated in a wide range of patients, including those with different types of glomerulonephritis (GN) evaluated longitudinally during treatment.⁹ However, the results may be influenced by reduced creatinine excretion because of low muscle mass. Thus, in elderly and female patients, PCR values can be higher than in young men. Some investigators consider a normal PCR sufficient to rule out pathologic proteinuria, but an elevated PCR should be confirmed and quantified with a 24-hour collection.¹² Others have found poor correlation between PCR and 24-hour proteinuria at high levels of protein excretion¹³ or that PCR is an unreliable method to monitor some patients with lupus nephritis.¹⁴

The KDIGO guideline suggests ACR rather than PCR as the first measurement of proteinuria in adults because albuminuria is a reliable marker of the outcome of CKD, and it provides a specific and sensitive measure of changes in glomerular permeability in several kidney diseases.¹

However, false-negative results may occur with ACR, especially in tubulointerstitial diseases and monoclonal gammopathies, in which urine proteins are mostly composed of tubular proteins and monoclonal light chains, respectively. In children, the KDIGO guideline recommends the measurement of PCR rather than ACR because the latter can miss the identification of congenital disorders associated with non-albumin proteinuria.¹ KDIGO also recommends PCR in preference to ACR for patients with glomerulonephritis.

Specific Proteins

Albumin. Albuminuria may be classified as mildly, moderately, or severely increased.¹ In persons with diabetes, abnormal albuminuria is correlated with a risk for progressive kidney function loss and is correlated among those with or without diabetes with increased risk for CKD, cardiovascular morbidity, and all-cause mortality.¹⁵ Semiquantitative reagent strips are available to screen for urine albumin in this range. Once the reagent strip is positive, a quantitative method must be used for confirmation.¹

Tubular proteins. When an isolated tubular lesion is suspected, specific tubular proteins such as α_1 -microglobulin, retinol-binding protein, or β_2 -microglobulin should be measured.¹ This can be done by qualitative analysis of urine proteins, using electrophoresis.

Bence Jones protein. Bence Jones proteinuria indicates the presence of free monoclonal immunoglobulin (heavy or light chains) as occurs with monoclonal gammopathies. Bence Jones proteinuria is revealed by urine electrophoresis, whereas light-chain identification requires urine immunofixation.¹⁶

Leukocyte Esterase

The leukocyte esterase dipstick test evaluates the presence of leukocytes based on the activity of an indoxyl esterase released from lysed

neutrophil granulocytes. Leukocyte esterase may be positive but microscopy negative when leukocytes are lysed because of low SG, alkaline pH, or a delay in sample handling and examination. False-negative results derive from vitamin C,⁷ high glucose (≥ 20 g/L), or high protein (≥ 5 g/L) concentration or from the presence of antibiotics, such as cephalothin and tetracycline (strong inhibition), cephalexin (moderate inhibition), or tobramycin (mild inhibition). The sensitivity is also reduced by high SG because this prevents leukocyte lysis. False-positive results may occur when formaldehyde is used as a urine preservative, from the presence in the urine of imipenem, meropenem, or clavulanate,¹⁷ and with all discolored urine.

Nitrites

The dipstick nitrites test detects bacteria that reduce nitrates to nitrites by nitrate reductase activity. This includes most gram-negative uropathogenic bacteria, but not *Pseudomonas*, *Staphylococcus albus*, or *Enterococcus*. False-negative results also may occur on a diet with low content of nitrate (vegetables), which form the substrate for nitrite production and short bladder incubation time. Thus, the sensitivity of the dipstick nitrites test is low, whereas specificity is high.¹⁸ False-positive results may occur in the presence of abnormally colored urine.

Bile Pigments

Measurement of urinary urobilinogen and bilirubin concentrations has lost its clinical value in the detection of liver disease after the introduction of serum tests of liver enzyme function.

Ketones

The ketone dipstick tests for acetoacetate and acetone (but not β -hydroxybutyrate), which are excreted into urine during diabetic acidosis or during fasting, vomiting, or strenuous exercise. It is based on the reaction of the ketones with nitroprusside.

URINE MICROSCOPY

In current days, urine microscopy is carried out almost exclusively in clinical laboratories. The lack of clinico-laboratory correlation⁴ may have an impact on the diagnosis and management of kidney patients. We believe urine microscopy remains a fundamental skill for nephrologists, as we will explore later.

Methods

A shared standardization for urine microscopy sample preparation and examination is lacking. We instruct the patient to deliver the second urine specimen of the morning because it avoids the lysis of particles that can occur in the bladder overnight (Box 4.1). In our laboratory we use the following standardized procedures for urine sample preparation and examination.⁶ We centrifuge an aliquot of urine within 3 hours from collection and concentrate it by removal of a fixed aliquot of supernatant urine. After this, the sediment is resuspended with a Pasteur pipette, and a fixed aliquot is transferred to the slide and prepared using a coverslip with a fixed surface. Some suggest the use of noncentrifuged urine because centrifugation may cause the damage and/or lysis of particles during the procedure. On the other hand, with this approach, clinically important particles (e.g., erythrocyte casts), when in small numbers, can easily be missed.

We recommend the use of phase-contrast microscopy because it improves the identification of almost all particles, especially cells and casts, whereas polarized light is mandatory for the correct identification of lipids and crystals, especially when they have uncommon morphologies.⁶

BOX 4.1 Standardized Procedures for Preparation and Examination of Urine Sediment Used in the Authors' Laboratory

- Written instructions for the patient to deliver a correct urine sample (i.e., the second urine of the morning after discarding the first portion of urine [midstream urine] collected in a proper container).
- Sample handling and examination within 3 hours of collection.
- Macroscopic examination of the sample and sediment.
- Testing the urine sample with a reagent strip for specific gravity, pH, albumin, hemoglobin, leukocyte esterase, and nitrites.
- Centrifugation of a 10-mL aliquot of urine at 400 g for 10 minutes.
- Removal of 9.5 mL of supernatant urine.
- Gentle but thorough resuspension with a Pasteur pipette of the sediment in the remaining 0.5 mL of urine.
- Transfer by a precision pipette of 50 μ L of resuspended urine to a glass slide.
- Covering of sample with a 24 $\times$ 32 mm glass coverslip.
- Examination of the urine sediment with a phase contrast microscope at low ($\times$ 160) and high magnification ($\times$ 400).
- Use of polarized light to identify doubtful lipids and crystals.
- Matching of the microscopic findings with reagent strip results.
- Cells expressed as lowest/highest number seen per high-power field, casts as number per low-power field, and all other elements (e.g., bacteria, crystals) on scale from 0 to ++++.

At least 20 microscopic fields, in different areas of the sample, should be examined at both low magnification (e.g., $\times$ 100 or $\times$ 200) and high magnification (e.g., $\times$ 400). More extensive examination may be required in certain clinical settings, such as isolated microhematuria of unknown origin, for which we suggest examination of 50 low-power fields (lpf) to look for erythrocyte casts.¹⁹

For correct examination, both pH and SG of the sample must be known. Both alkaline pH ($\geq$ 7.0) and low SG (especially $\leq$ 1.010) favor the lysis of erythrocytes and leukocytes, which can cause discrepancies between dipstick readings and the microscopic examination (see earlier discussion). Alkaline pH also impairs the formation of casts and favors the precipitation of amorphous phosphates. On the contrary, high SG ($\geq$ 1.030) may reduce the sensitivity of reagent strips for hemoglobin and leukocyte esterase.

We quantify the particles seen as number per microscopic field, whereas if counting chambers are used, the elements are quantified as number per volume.

Cells

Erythrocytes

Urinary erythrocytes have a mean diameter of approximately 6 μ m. In the urine, there are two main types of erythrocytes: *isomorphic*, with regular shapes and contours, derived from the urinary excretory system or from nonglomerular kidney diseases; and *dysmorphic*, with irregular shapes and contours, which are of glomerular origin (see Fig. 4.1A–B).²⁰ Erythrocyte dysmorphism is thought to result from deformation of the erythrocytes as they pass through gaps in the glomerular basement membrane, followed by physicochemical insults when the erythrocytes pass through the tubular system.²¹

The distinction between glomerular and nonglomerular hematuria is of special value in the evaluation of patients with isolated microhematuria, in whom it is important to decide whether nephrologic or urologic investigation is needed.

Unfortunately, there is no agreement on the criteria to classify hematuria as glomerular or nonglomerular. Some define glomerular

hematuria as more than 80% of erythrocytes being dysmorphic; others define the discriminating cut-off as low as 10% or 15%.⁶ Still, others define hematuria as glomerular when at least 5% of erythrocytes examined are *acanthocytes*,²² a subtype of dysmorphic erythrocytes with a distinguishing appearance easily identifiable by the presence of one or more blebs of different size and shape protruding from a ring-shaped body (see Fig. 4.1B, *inset*).

In our laboratory, glomerular hematuria is diagnosed when there are 40% or more dysmorphic erythrocytes and/or 5% or more acanthocytes and/or one or more red blood cell casts/50 lpf ($\times$ 160). With this approach, a good correlation was found between urinary sediment and kidney biopsy findings in 16 patients with long-standing isolated microhematuria.¹⁹ Rare types of erythrocytes found in the urine include sickle cells, elliptocytes, spherocytes, and dacryocytes.²³

Leukocytes

Urinary *neutrophils* have an average diameter of approximately 10 μ m and are the most frequently found leukocytes in the urine. Neutrophils are identified by their granular cytoplasm and lobulated nucleus (see Fig. 4.1C). In most patients, neutrophils indicate UTI, but they may also result from urine contamination caused by genital secretions. Variable numbers of neutrophils are often found in acute interstitial nephritis (AIN).²⁴ Neutrophils can be found in low numbers in chronic interstitial nephritis and in proliferative GN.²⁵

Eosinophils, which can be identified only by the use of stains (e.g., Hansel), were once considered a marker of acute allergic interstitial nephritis. This is not specific, however, because eosinophils may be present in various types of GN, prostatitis, chronic pyelonephritis, urinary schistosomiasis, and cholesterol embolism.²⁶

Lymphocytes, whose identification also requires staining, may indicate acute cellular rejection in kidney allograft recipients, although this is not sufficiently reliable to preclude the need for kidney biopsy. Lymphocytes are also a typical finding in patients with chyluria.

Macrophages are mononucleated or multinucleated cells of variable size (13–95 μ m in diameter) and variable appearance, with some being granular (see Fig. 4.1D). In patients with nephrotic syndrome, macrophages may be engorged with lipid drops, appearing as oval fat bodies (see lipids). Macrophages have been found in the urine of patients with active GN. In our experience, macrophages are frequently seen in the urine of kidney transplant recipients with BK virus infection (see later discussion). However, urinary macrophages are not diagnostic of any specific condition.

Renal Tubular Epithelial Cells

The renal tubular epithelial cells (RTECs) derive from the exfoliation of the tubular epithelium. In the urine, RTECs can differ in size (diameter approximately 9–25 μ m, average 14 μ m) and shape, from roundish to rectangular or columnar, with a central or peripheral large nucleus (see Fig. 4.1E). RTECs are not found in the normal individual but can be found when there is acute tubular damage, including ATN,²⁷ AIN,²⁴ and acute cellular rejection. In smaller numbers, RTECs also can be found in glomerular diseases.²⁵ In ATN, these cells are frequently damaged and necrotic and may be present in casts (so-called epithelial casts).

Transitional Epithelial Cells

The transitional epithelial cells derive from the exfoliation of the uroepithelium, which lines the urinary tract from the calyces to the bladder in women and to the proximal urethra in men. This multilayered epithelium has small cells in the deep layers and larger cells in the superficial layers. When cells of the deep epithelial layers (average diameter 18 μ m; see Fig. 4.1F) are present in large numbers (e.g., $\geq$ 1/high-power field [hpf]), this suggests severe uroepithelial damage, such as caused

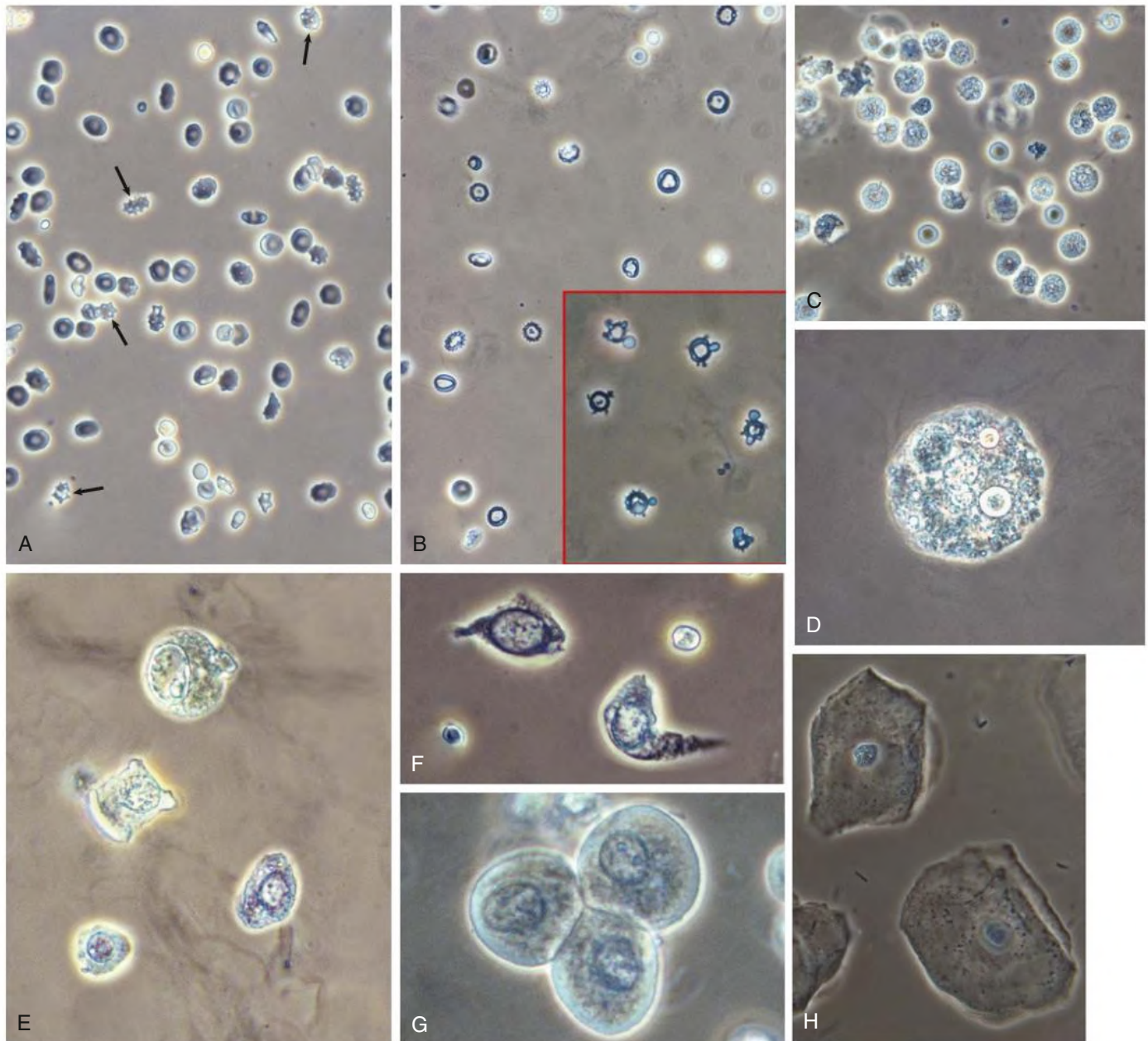


Fig. 4.1 Urinary Sediment Cells. (A) Isomorphic nonglomerular erythrocytes (diameter $\sim 6\ \mu\text{m}$). The arrows indicate the so-called crenated erythrocytes, which are a finding in nonglomerular hematuria. (B) Dysmorphic glomerular erythrocytes (diameter $\sim 6\ \mu\text{m}$). The dysmorphism consists mainly of irregularities of the cell membrane. *Inset*, Acanthocytes, with their typical ring-formed cell bodies with one or more blebs of different sizes and shapes. (C) Neutrophils (diameter $\sim 10\ \mu\text{m}$). Note their typical lobulated nucleus and granular cytoplasm. (D) Granular phagocytic macrophage (diameter $\sim 60\ \mu\text{m}$). (E) Different types of renal tubular epithelial cells (diameter $\sim 14\ \mu\text{m}$). (F) Two cells from deep layers of uroepithelium (diameter $\sim 18\ \mu\text{m}$). (G) Three cells from superficial layers of uroepithelium (diameter $\sim 25\ \mu\text{m}$). (Note the difference in shape, size, and ratio of nucleus to cytoplasm between the two types of uroepithelial cells.) (H) Squamous epithelial cells (diameter $\sim 50\ \mu\text{m}$). (All images by phase-contrast microscopy; original magnification $\times 400$.)

by neoplasia, stones, obstruction, or long-standing bladder catheters or ureteral stents.⁶ Transitional cells of the superficial layers (average diameter $\sim 25\ \mu\text{m}$; see Fig. 4.1G) are a common finding associated with mild uroepithelial damage, as may occur in cystitis.

Squamous Epithelial Cells

Squamous epithelial cells (SECs; average diameter $50\ \mu\text{m}$; see Fig. 4.1H) derive from the urethra or from the external genitalia. In small

numbers, SECs are a normal finding, but in large numbers, they indicate urine contamination from genital secretions and reflect improper urine collection technique.

Lipids

Lipids are found in the urine as *drops*, which are spherical, translucent, yellowish particles of different sizes that can be isolated or in clusters, free or within cells or casts; as *oval fat bodies*, which are RTECs or

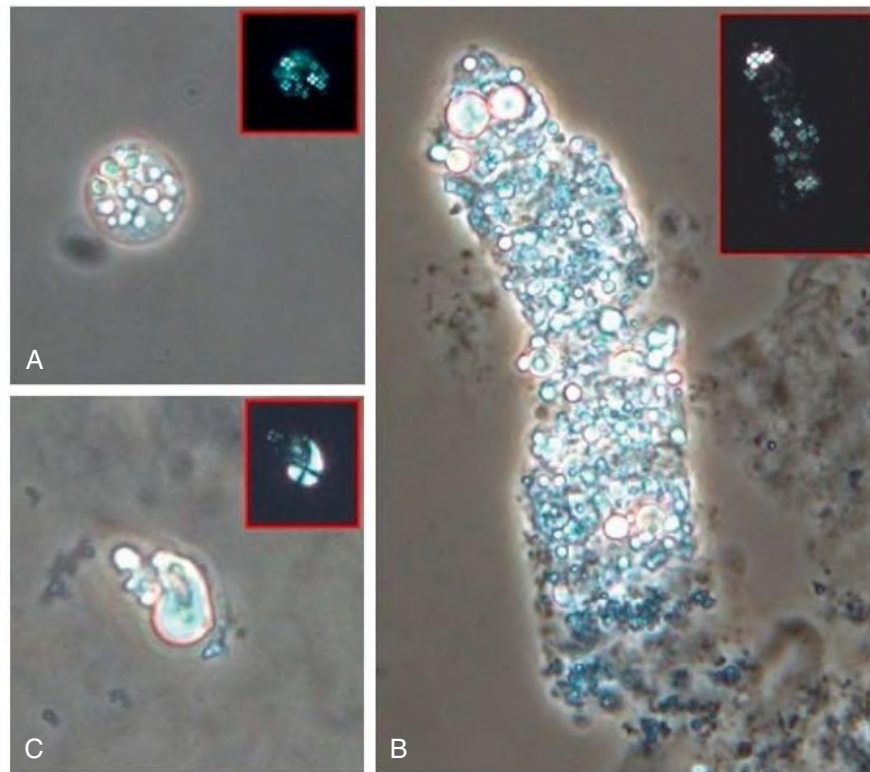


Fig. 4.2 Fat Particles. (A) Oval fat body and (B) fat cast—containing lipid drops with very variable diameter which, under polarized light, show a symmetric “Maltese cross” birefringence (*insets*). (C) Fat particle with an ovoid shape and multiple blebs as found in the urine of patients with Fabry disease which, under polarized light, shows a truncated and asymmetric Maltese cross appearance (*inset*) (Phase-contrast microscopy, original magnification $\times 400$.)

macrophages gorged with lipid drops (Fig. 4.2A); as *fat casts*, cylindrical structures containing variable amounts of fat drops, or even oval fat bodies (see Fig. 4.2B); and *cholesterol crystals* (see Crystals and Fig. 4.4F). All these particles contain mainly cholesterol esters and free cholesterol. Under polarized light, lipids are birefringent and show “Maltese crosses” with symmetric arms (see Fig. 4.2A and 4.2B *inset*), whereas cholesterol crystals are nonbirefringent. These lipids are typical of glomerular diseases associated with marked proteinuria.

In Fabry disease, urine sediment may contain fat particles even in the absence of proteinuria. These particles contain glycosphingolipids (especially globotriaosylceramide-3) and differ from the fat particles previously described because they show quite variable shapes, often with external protrusions and/or an internal lamellar structure, and irregular or truncated Maltese crosses under polarized light (see Fig. 4.2C).²⁸

Casts

Casts are cylindrical structures that form in the lumen of distal renal tubules and collecting ducts. Their matrix is made of Tamm-Horsfall glycoprotein, today known as uromodulin, which is secreted by the cells of the thick ascending limb of Henle loop. Trapping of particles within the cast matrix results in casts with different appearances, each of which may have specific clinical significance (Table 4.3). It is important to remember that because casts form in the renal tubules, whatever particle is contained in a cast derives from the kidneys. Specific casts include the following:

- *Hyaline casts* are colorless with a low refractive index (Fig. 4.3A). They are easily seen with phase-contrast microscopy but can be overlooked when bright-field microscopy is used. Hyaline casts can

be found both in normal and pathologic urine and especially in the setting of dehydration.

- *Hyaline-granular casts* contain variable amounts of granules within the hyaline matrix (see Fig. 4.3B) and are the most common mixed casts (see later discussion). Hyaline-granular casts can occasionally be found in normal subjects. They are common in patients with kidney diseases such as GN and AIN, in which case they are associated with other types of casts.^{24,25}
- *Granular casts* can be finely (see Fig. 4.3C) or coarsely granular. Both types indicate kidney disease. In patients with acute kidney injury (AKI), granular casts, together with RTECs²⁷ or epithelial casts,²⁹ are a sensitive marker of ATN.
- *Waxy casts* derive their name from their appearance, which is similar to that of melted wax (see Fig. 4.3D). They are typically found in patients with kidney disease associated with reduced GFR, whether acute, rapidly progressive, or chronic.³⁰
- *Fat casts* contain variable amounts of lipid drops, isolated, in clumps, or even oval fat bodies or cholesterol crystals (see Fig. 4.2B). Fat casts are typical of glomerular diseases associated with marked proteinuria.
- *Erythrocyte casts* may contain a few erythrocytes (see Fig. 4.3E) or so many that the matrix of the cast cannot be identified. Erythrocyte casts are a marker of *renal bleeding*, most frequently of glomerular origin, even though they have also been found in patients with AIN.²⁴
- *Hemoglobin casts* generally have a brownish hue and a coarsely granular appearance, which derives from the degradation of erythrocytes entrapped within the cast matrix (see Fig. 4.3F). In such cases, hemoglobin casts have the same clinical significance as

TABLE 4.3 Types of Casts and Their Main Clinical Associations

Cast	Main Clinical Associations
Hyaline	Normal individual; kidney disease
Hyaline-granular	Normal subject; kidney disease
Granular	Kidney disease; AKI associated with ATN
Waxy	Kidney disease with possible functional impairment
Fat	Marked proteinuria, usually but not invariably in the nephrotic range
Erythrocyte	Glomerular hematuria, most frequently because of active proliferative GN; AIN
Leukocyte	AIN; acute pyelonephritis; active proliferative GN
Containing renal tubular epithelial cell (so-called epithelial casts)	AKI associated with ATN; active proliferative GN; nonproliferative GN associated with heavy proteinuria; AIN
Hemoglobin	Glomerular hematuria, most frequently because of active proliferative GN; hemoglobinuria caused by intravascular hemolysis
Myoglobin	AKI because of rhabdomyolysis
Bilirubin	Jaundice caused by increased direct bilirubin
Bacterial, fungal	Bacterial or fungal kidney infections
Crystalline	Crystalline nephropathies
Mixed	According to components present in the cast

AIN, Acute interstitial nephritis; AKI, acute kidney injury; ATN, acute tubular necrosis; GN, glomerulonephritis.

erythrocyte casts. However, hemoglobin casts also may derive from hemoglobinuria, as may occur in intravascular hemolysis. In these patients, hemoglobin casts usually have a smooth surface.

- *Leukocyte casts* contain variable amounts of polymorphonuclear leukocytes (see Fig. 4.3G). They can be found in patients with acute pyelonephritis and AIN²⁴ and in active proliferative GN.²⁵
- *Renal tubular epithelial cell containing casts* (so-called epithelial casts) contain variable numbers of RTECs, which can be identified by their prominent nucleus (see Fig. 4.3H). Epithelial casts indicate damage of the renal tubular epithelium and can therefore be found in the urine of patients with ATN,²⁷ AIN,²⁴ and glomerular disease.²⁵
- *Myoglobin casts* are cylinders pigmented by myoglobin, akin to hemoglobin casts (see Fig. 4.3F). They can be distinguished from myoglobin casts according only to the clinical setting. Myoglobin casts are observed in the urine of patients with AKI associated with rhabdomyolysis.
- *Bilirubin casts* are cylinders pigmented with bilirubin, which can stain any particle contained in the cast (see Fig. 4.3I). They are observed in the urine of patients with jaundice associated with increased direct (conjugated) bilirubin.
- *Crystalline casts* may contain crystals of different nature. These casts are an important diagnostic element in crystalline-induced nephropathies, which are usually associated with AKI.³¹
- *Casts containing microorganisms* (bacteria and yeasts) indicate a kidney infection.
- *Mixed casts* contain components of different nature, such as granules, cells, and lipids. Their clinical significance is the same as that of pure casts.

Crystals

Correct identification of urine crystals requires the combined knowledge of their most frequent morphologies, their birefringence features under polarized light, and urine pH range at which each type of crystals precipitates. For unusual crystals, however, additional investigation may be needed, such as infrared or Raman spectroscopy and other techniques that are available in specialized laboratories.³²

Examination of the urine for crystals is a key test in the assessment of patients with stone disease, with some rare inherited metabolic disorders (e.g., cystinuria, hyperoxaluria, phosphoribosyltransferase deficiency), and with suspected drug nephrotoxicity.⁶ The search for crystalluria is best performed on the first voided morning urine samples, which are the most concentrated. We classify crystals into four categories: common crystals, pathologic crystals, rare crystals caused by drugs, and other crystals.

Common Crystals

Uric acid crystals and amorphous urates. These crystals most commonly appear as rhomboidal or barrel-like structures with a typical yellow amber color (Fig. 4.4A). However, uric acid crystals may have a wide spectrum of appearances, such as needle-like or grenade-like morphology.^{33,34} Under polarized light, uric acid crystals are strongly birefringent and polychromatic. They are found in acidic urine (pH 5.0–5.8).

Amorphous urates are tiny granules of irregular shape that polarize light and precipitate in acidic urine.

Calcium oxalate crystals. There are two types of calcium oxalate crystals: bihydrated (or weddellite) crystals, which most often have a bipyramidal appearance (see Fig. 4.4B), and monohydrated (or whewellite) crystals, which are ovoid, dumbbell-shaped, or biconcave disks (see Fig. 4.4C). Monohydrated crystals are birefringent under polarized light, whereas bihydrated crystals in most instances are not. Both types precipitate at pH 5.4 to 6.7.

Calcium phosphate crystals (brushite) and amorphous phosphates. Calcium phosphate crystals are pleomorphic, appearing as prisms, star-like particles, or needles of various sizes and shapes (see Fig. 4.4D) that are strongly birefringent under polarized light. Rarely, calcium phosphate can appear as plates with a granular surface that do not polarize light. Both types of crystals precipitate in alkaline urine (pH $\geq$ 7.0).

Amorphous phosphates are tiny particles identical to amorphous urates, but they do not polarize light and precipitate at a pH of 7.0 or higher.

Triple phosphate (struvite) crystals. These crystals contain magnesium ammonium phosphate and most frequently have the appearance of “coffin lids” (see Fig. 4.4E), although variants such as flower-like and scissors-like structures can be found. These crystals usually polarize light strongly and are found in alkaline urine (pH $\geq$ 7.0).

Pathologic Crystals

Cholesterol crystals. These are thin, transparent plates, often clumped together, with sharp edges (see Fig. 4.4F), which do not polarize light. They can be found at any urinary pH.

Cystine crystals. These are hexagonal plates with irregular sides that are often heaped on one another (see Fig. 4.4G). They either do not polarize light or show a whitish birefringence. They are insoluble in a urine pH up to 7.4. Their presence is pathognomonic of the autosomal recessive inherited condition cystinuria (see Chapter 50), and their persistence in urine samples indicates a high risk for urinary stone formation.³⁵



Fig. 4.3 Casts. (A) Hyaline cast. (B) Hyaline-granular cast. (C) Finely granular cast. (D) Waxy cast. (E) Erythrocyte cast. (F) Hemoglobin cast with a coarsely granular appearance and typical brownish hue. (G) Leukocyte cast. Note the lobulated nucleus of polymorphonuclear leukocytes (*arrows*). (H) Epithelial cell cast. (Note, in its lower extremity, the large nucleus of the renal tubular epithelial cells.) (I) Bilirubin cast with a coarsely granular appearance and typical yellow color. (All images by phase-contrast microscopy; original magnification $\times 400$.)

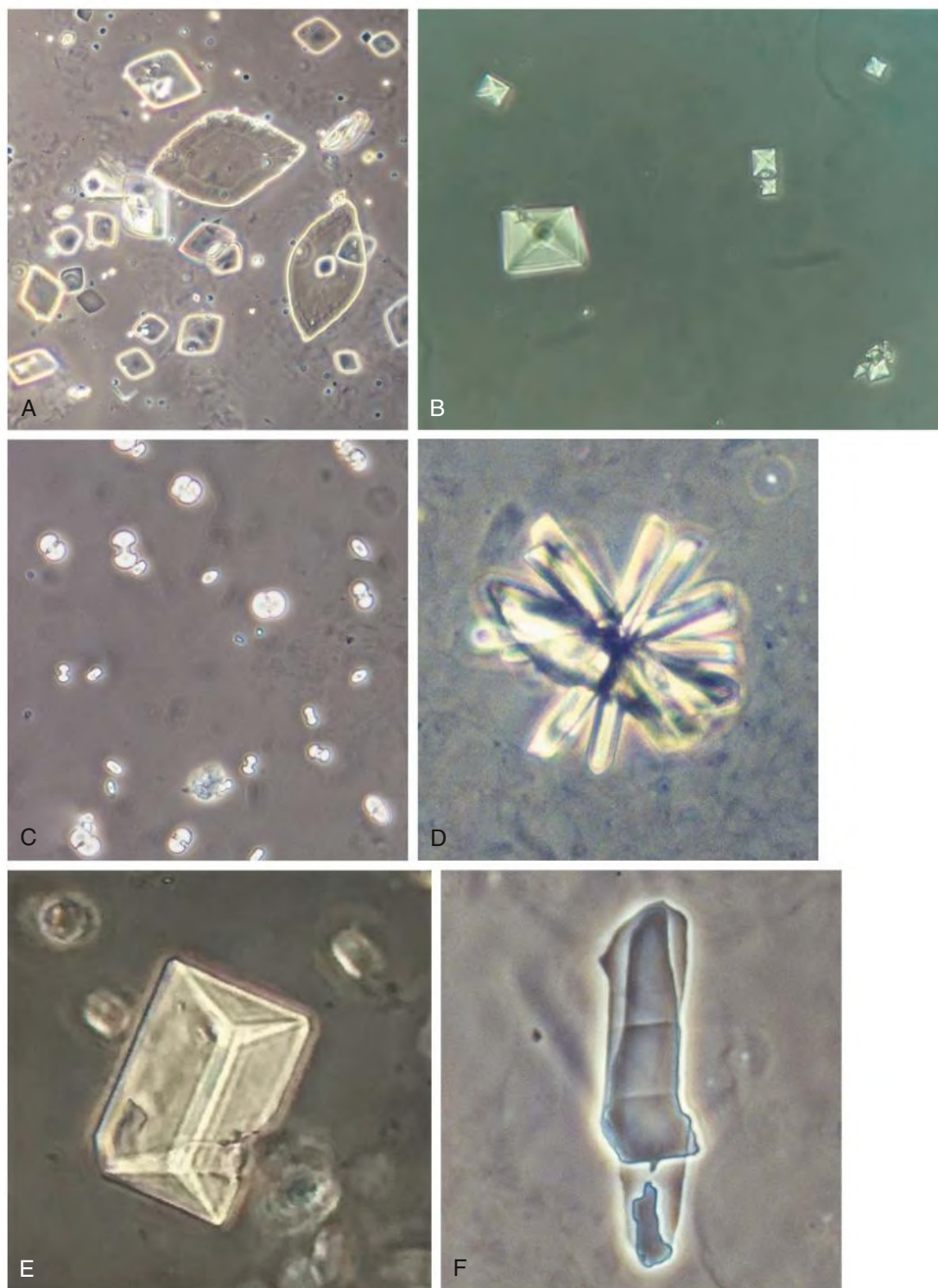


Fig. 4.4 Crystals. (A) Uric acid crystals. This rhomboid shape is the most common. (B) Bihydrated calcium oxalate crystals. They appear as bipyramidal colorless structures of variable size. (C) Different types of monohydrated calcium oxalate crystals. (D) Star-like brushite (calcium phosphate) crystal. (E) Struvite (triple phosphate) crystal, with typical “coffin-lid” appearance. (F) Cholesterol crystal.

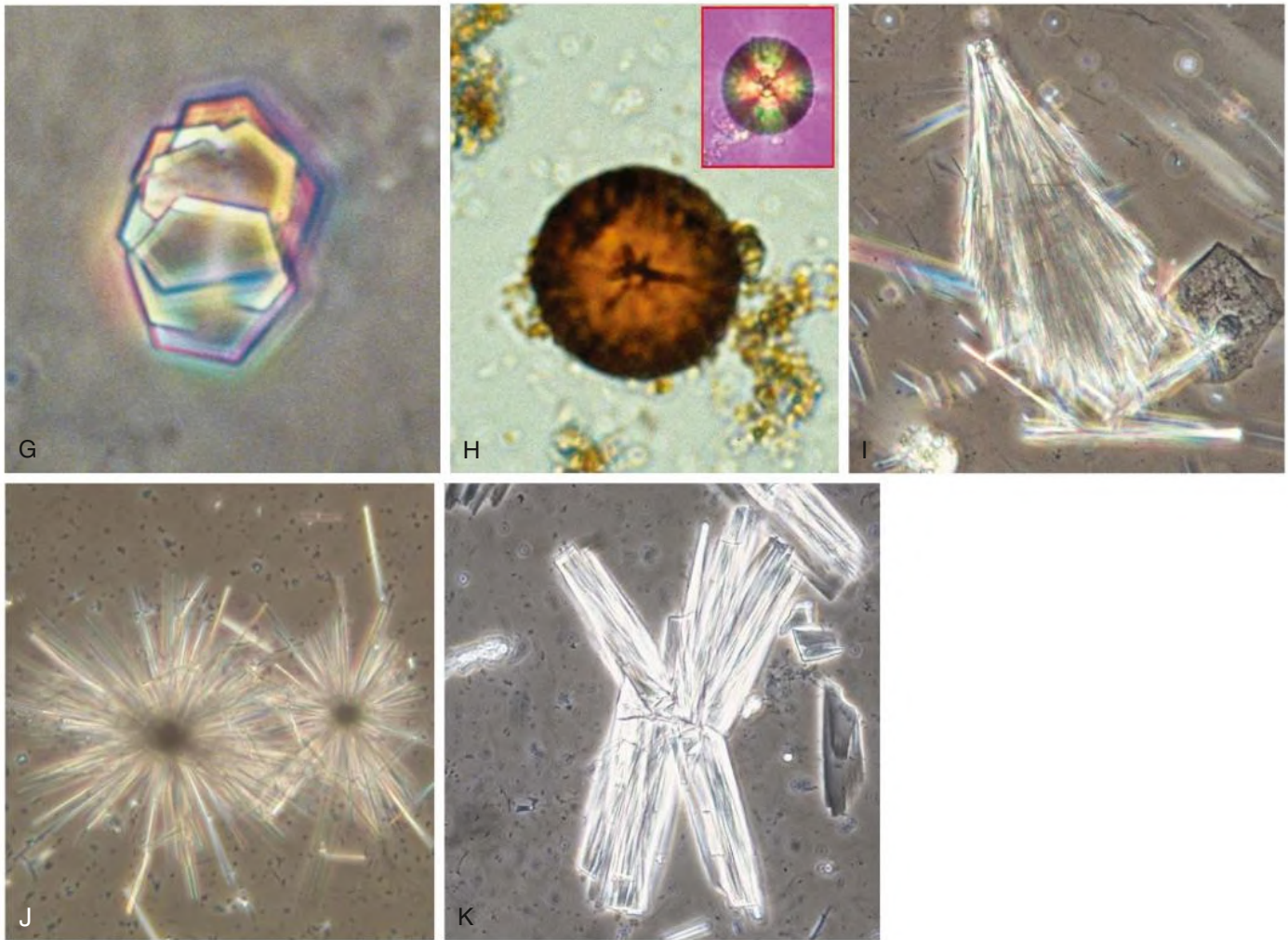


Fig. 4.4, cont'd (G) Cystine crystals heaped one on the other. (H) 2,8-Dihydroxyadenine crystal by bright-field microscopy; *inset*, by polarized light. (I) Amoxicillin crystal resembling a broom or brush. (J) Star-like ciprofloxacin crystals. (K) Large crystal of indinavir. (All images by phase-contrast microscopy; original magnification $\times 400$.) (H, Courtesy Doctor Michel Daudon, Paris.)

2,8-dihydroxyadenine (2,8-DHA) crystals. These are spherical brownish structures with a central umbilicus and a birefringent cross-like appearance under polarized light (see Fig. 4.4H). They are a marker of homozygous deficiency of the enzyme adenine phosphoribosyltransferase.³⁶

Tyrosine and leucine crystals. Both crystals are rare and can be found in the urine of patients with severe liver disease. In addition, tyrosine crystals may be found in patients with tyrosinemia. Leucine crystals appear as yellow-brown spheres with concentric striations, which are birefringent under polarized light. Tyrosine crystals are needle-shaped and often aggregate.

Crystals Caused by Drugs

Several drugs can cause crystalluria. This happens in the presence of several factors that increase drug supersaturation, such as drug overdose, dehydration, hypoalbuminemia, or reduced kidney function, in the setting of a crystallization favoring pH range (which varies from drug to drug).

Drugs that can cause crystalluria include antimicrobials (sulfadiazine, amoxicillin [see Fig. 4.4I], ciprofloxacin⁶ [see Fig. 4.4J], ceftriaxone, and sulfamethoxazole³⁷); antivirals (acyclovir, indinavir⁶ [see Fig. 4.4K], atazanavir, and darunavir³⁸); and other drugs (such as

naftidrofuryl oxalate, orlistat, intravenous vitamin C, primidone, felbamate, piridoxylate, sulfasalazine, and methotrexate).^{3,39}

In the presence of crystals with morphologies that differ from those previously described, a drug-related etiology must be sought. Calcium oxalate crystals of drug-related origin (naftidrofuryl oxalate, orlistat, and vitamin C), however, are indistinguishable from other causes.⁶

Other Crystals

Hippuric acid crystals, calcium carbonate crystals, and ammonium biurate crystals are rare, and it is unclear whether they are clinically significant.

Clinical Significance of Crystals

Uric acid, calcium oxalate, and calcium phosphate crystals may have no clinical significance because they can reflect transient supersaturation of the urine caused by ingestion of some foods (e.g., meat for uric acid, spinach or chocolate for calcium oxalate, milk or cheese for calcium phosphate) or mild dehydration. However, the persistence of calcium oxalate or uric acid crystalluria may reflect hypercalciuria, hyperoxaluria, or hyperuricosuria. In calcium stone formers, the evaluation of crystalluria is an important tool to assess calcium stone disease activity.⁴⁰

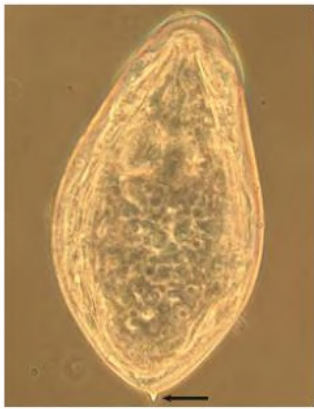


Fig. 4.5 Egg of *Schistosoma haematobium* (Diameter, ~100 μ m). Note the thick shell, which contains the *miracidium*, and the typical terminal spike (arrow). (Phase-contrast microscopy; original magnification $\times 400$.)

Large numbers of uric acid crystals may be associated with AKI caused by acute urate nephropathy, whereas large numbers of monohydrated calcium oxalate crystals, especially with a spindle shape, may be associated with AKI from ethylene glycol intoxication. Triple phosphate crystals are usually associated with UTI caused by urea-splitting microorganisms such as *Proteus* spp., *Ureaplasma urealyticum*, and *Corynebacterium urealyticum*.

Cholesterol crystals are found in association with other fat particles in patients with marked proteinuria. Cystine crystals are pathognomonic of cystinuria, and 2,8-dihydroxyadenine crystals are pathognomonic of adenine phosphoribosyltransferase enzyme deficiency. Crystalluria may be isolated and asymptomatic or associated with hematuria, obstructive uropathy, or AKI caused by the precipitation of crystals within the renal tubules.^{6,31}

Organisms

Bacteria: Two types of bacteria are most frequently found in urine—cocci, which have a round to oval shape, and rods, which appear as short sticks. High numbers of bacteria in association with leukocytes are a marker of urinary tract infection. However, the copresence of high numbers of squamous epithelial cells and/or *Candida albicans* and/or *Trichomonas vaginalis* suggests urine contamination from genital secretions.

Candida albicans: It is the most frequent yeast found in urine. Its shape is ovoidal, with one or two round to elongated protruding buds of variable size.

Trichomonas vaginalis: It is a protozoon characterized by a pyriform body, the presence of flagella at the two extremities of the body, and quick and whirling movements through the slide.

Schistosoma haematobium: It is an endemic parasite in several geographic regions (see Chapter 56). The search in the urinary sediment of its eggs is the simplest and fastest diagnostic tool. Their distinguishing feature is a terminal spike (Fig. 4.5). The eggs are most frequently found in the urine collected between 10 AM and 2 PM, when the parasite female lays the eggs, and after physical exercise such as running, which favors the detachment of the eggs from the bladder mucosa.

Contaminants

A large number of particles can contaminate urine. These particles may come from the patient (e.g., spermatozoa; erythrocytes from menstruation; leukocytes from vaginitis, cloth or synthetic fibers from underclothes), the laboratory (e.g., glass fragments from coverslips, starch

from gloves), or the environment (e.g., pollens, plant cells, fungal spores).⁶

INTERPRETATION OF URINE SEDIMENT FINDINGS

Examination of the urine sediment, coupled with the knowledge of proteinuria and other urine and blood findings, results in urine sediment profiles that aid in diagnosis of urinary tract diseases (Table 4.4).

Acute Kidney Injury

In patients with AKI, the finding in the urine sediment of RTECs, in association with granular casts and/or epithelial casts, is the hallmark of ATN,^{27,29} whereas these elements are rarely found in prerenal AKI.²⁷ In the latter, hyaline and/or hyaline-granular casts may be seen, along with high SG and low pH. A score based on the number of RTECs and granular casts significantly correlates with the progression and severity of AKI, with new AKI urine biomarkers (NGAL, KIM-1, IL 18), with the need for dialysis, and with death.²⁷ Depending on the cause of the tubular damage, other elements can be seen (e.g., high numbers of dysmorphic erythrocytes and erythrocyte casts in active proliferative glomerular diseases; myoglobin-pigmented casts in AKI because of rhabdomyolysis; and massive amounts of uric acid crystals in acute uric acid nephropathy caused by tumor lysis syndrome).

Acute Interstitial Nephritis

In AIN, the most frequent findings are variable numbers of leukocytes and isomorphic erythrocytes. RTECs may also be present, together with epithelial, leukocyte, and/or erythrocyte casts.²⁴

Active Proliferative Glomerulonephritis

Dysmorphic erythrocytes and erythrocyte and hemoglobin casts are the hallmark of active proliferative glomerulonephritis. Usually, the number of erythrocytes ranges from 30 to 40 cells/hpf to more than 100 cells/hpf, with the higher figure found especially in patients with extracapillary or necrotizing glomerular lesions. Leukocytes are also common in mild numbers (e.g., 3–5/hpf) in most patients, but in those with acute postinfectious GN or active proliferative lupus nephritis, we have seen samples with up to 30 to 40 leukocytes/hpf. Leukocyte casts and waxy casts³⁰ may also be observed. In patients with marked proteinuria, fat particles can be also seen.

Nephrotic Syndrome

The typical nephrotic sediment contains lipids, casts, and RTECs. Fat, epithelial, granular, hyaline, and hyaline-granular casts are frequent, whereas erythrocyte or hemoglobin casts, leukocyte casts, and waxy casts are few or absent. Erythrocytes may be totally absent, especially in minimal change disease or may be in low to moderate numbers (e.g., 3–5/hpf to 20–30/hpf), which is seen especially in membranous nephropathy and focal segmental glomerulosclerosis. Leukocytes are usually not found.

Urinary Tract Infection

Bacteria and leukocytes are the hallmarks of UTI, with or without superficial transitional epithelial cells and/or isomorphic erythrocytes. In patients with pyelonephritis, RTECs and leukocyte casts can also be found. When the infection is caused by urease-producing bacteria such as *Proteus* spp., *Ureaplasma urealyticum*, and *Corynebacterium urealyticum*, triple phosphate (struvite) crystals are often present. The finding of bacteria and leukocytes, together with a high number of squamous epithelial cells, suggests the presence of urine contamination from genital secretion rather than UTI.

TABLE 4.4 Main Urinary Profiles

Kidney Disease	Hallmark	Associated Findings
AKI with ATN (proteinuria: absent to +)	RTECs Epithelial casts Granular casts	Variable according to cause of ATN (e.g., high number of dysmorphic erythrocytes in active proliferative GN; myoglobin casts in rhabdomyolysis; uric acid crystals in acute urate nephropathy)
Acute interstitial nephritis (proteinuria: absent to +)	Leukocytes Isomorphic erythrocytes	RTECs Leukocyte casts Erythrocyte casts
Active proliferative glomerulonephritis (proteinuria: + to ++++)	Dysmorphic erythrocytes (moderate to high number) Erythrocyte/hemoglobin casts	Leukocytes (low to moderate number) RTECs Epithelial, hyaline, hyaline-granular, granular, waxy, leukocytic casts
Nephrotic syndrome (proteinuria: ++++)	Fat particles High number of casts (fat, hyaline, hyaline-granular, granular epithelial)	Renal tubular epithelial cells Dysmorphic erythrocytes (absent or low-to-moderate number)
Urinary tract infection (proteinuria: absent)	Bacteria Leukocytes	Isomorphic erythrocytes Transitional epithelial cells Triple phosphate crystals (for infections caused by urease-producing bacteria) Epithelial ± leukocyte casts (in pyelonephritis)
Polyomavirus BK infection (proteinuria: absent)	Decoy cells	Decoy cell casts (in BK virus nephropathy) Macrophages
Urologic diseases (proteinuria: absent)	Isomorphic erythrocytes (low to high number) Leukocytes Transitional cells (deep, superficial, atypical)	Crystals (urolithiasis) Atypical/malignant transitional cell (neoplasia of excretory urinary system)

AKI, Acute kidney injury; ATN, acute tubular necrosis; GN, glomerulonephritis; RTEC, renal tubular epithelial cell.

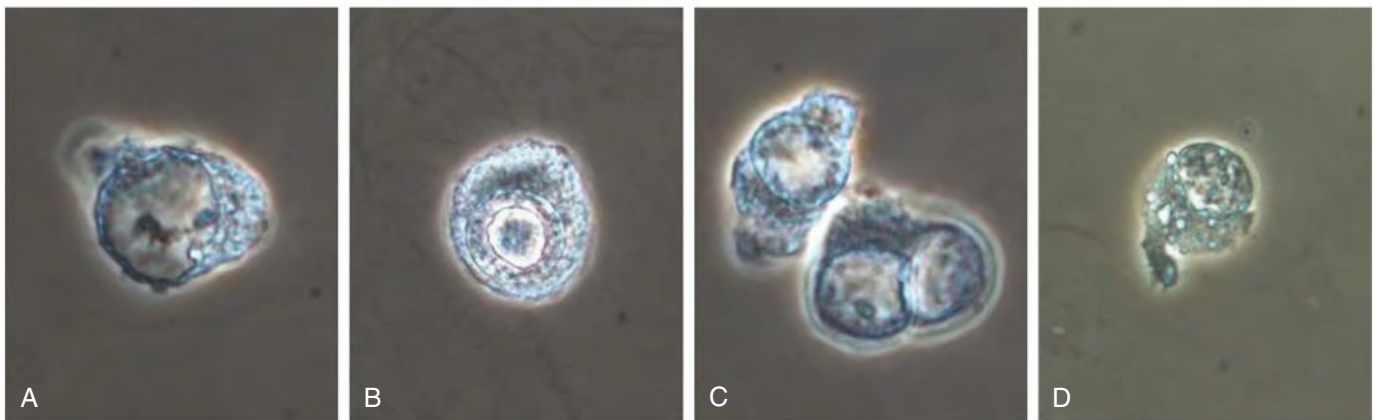


Fig. 4.6 Decoy Cells. (A) Cell with nuclear ground-glass or gelatinous appearance (phenotype 1). (B) Cytomegalovirus-like cell with a large intranuclear inclusion surrounded by a clear halo (phenotype 2). (C) A binucleated cell (bottom, phenotype 3) and a cell with an enlarged ground glass nucleus (phenotype 1). (D) Cell with clumped chromatin (phenotype 4). (Phase-contrast microscopy; original magnification $\times 400$.)

BK Virus Infection

The KDIGO clinical practice guideline for kidney transplant recipients recommends monitoring for BK polyomavirus (BKV) reactivation (which may lead to BKV nephropathy [BKVN] and graft loss) by measuring viral nucleic acid in the blood (i.e., viremia).⁴¹ This approach is expensive and not always available.⁴² As an alternative, searching for “decoy cells” on either smeared or cytocentrifuged alcohol-fixed and Papanicolaou-stained urine specimens provides satisfactory diagnostic accuracy.^{42,43}

However, decoy cells can also be easily identified by phase-contrast microscopy in routine unstained samples⁴⁴ (Fig. 4.6). Four decoy cell phenotypes are recognized⁴⁵: (1) nuclear ground-glass or gelatinous appearance (see Fig. 4.6A); (2) intranuclear inclusion surrounded by a clear halo (cytomegalovirus-like; see Fig. 4.6B); (3) multinucleated cells (see Fig. 4.6C); and (4) vesicular nuclei with clumped chromatin and nucleoli (see Fig. 4.6D). In addition, hybrid forms are frequently seen, as are cells with eccentric nucleus and comet-like appearance. The presence of decoy cells may indicate solely the reactivation of BKV

infection; however, when they persist over time, are in high numbers, or are found in urinary casts, they are a reliable marker of likely BKVN, which should be confirmed by measurement of BK viremia.^{42,43,45}

Urologic Diseases

Urinary tract disorders such as cancer, urolithiasis, and hydronephrosis are associated with variable numbers of isomorphic urinary erythrocytes, which are often associated with leukocytes or transitional epithelial cells (from deep or superficial layers of uroepithelium). In addition, in uroepithelial cancer, malignant transitional cells can be found, which show abnormal size and shape, increased number and size of nuclei, and enlarged nucleoli.⁴⁶

Nonspecific Urinary Abnormalities

Some urine sediment findings are nonspecific. This occurs when variable numbers of hyaline or hyaline-granular casts are found with or without low numbers of isomorphic erythrocytes, leukocytes, common crystals, or superficial transitional epithelial cells. In such cases, the correct interpretation of the urinary findings requires adequate clinical information and the knowledge of other laboratory tests.

AUTOMATED ANALYSIS OF URINE SEDIMENT

In high-income countries, automated urinary sediment analyzers are increasingly used in clinical laboratories where several hundreds of

urine samples are analyzed every day. Three main types of automated instruments using different technology are available: flow cytometry, automated intelligent microscopy, and cuvette-based microscopy.

Flow cytometry supplies quantitative results and graphics (“scattergrams”) of the identified particles.⁴⁷

Intelligent microscopy supplies quantitative results and images of the particles present in the sample, which are pooled and shown on the screen by particle categories (e.g., all erythrocytes, leukocytes).⁴⁸

Cuvette-based microscopy supplies high-quality grayscale images of elements within whole microscopic fields, obtained with both bright-field and phase-contrast microscopy.⁴⁹

Compared with manual microscopy, automated urine sediment analyzers offer several advantages: small volumes of urine are required (1–3 mL); high numbers of samples can be examined in a short time (up to 130/h); elimination of the drawbacks caused by sample centrifugation (e.g., time consumption, loss/lysis of particles); and quantitative results with small variation coefficients. However, they also have limitations: their findings correlate well with manual microscopy for elements such as erythrocytes, leukocytes, and SECs, but not for others such as casts and nonsquamous epithelial cells (i.e., RTECs and deep and superficial transitional cells). Finally, they cannot identify the subtypes of pathologic casts, lipids, decoy cells, and malignant atypical epithelial cells.

For these reasons, in spite of their continuous diagnostic improvement, automated instruments cannot yet replace manual microscopy for samples from renal patients.⁵⁰

SELF-ASSESSMENT QUESTIONS

1. The reagent strip for protein:
 - A. detects exclusively urinary albumin.
 - B. detects Bence Jones proteinuria.
 - C. is not influenced by the pH of the urine.
 - D. is not influenced by the specific gravity of the urine.
2. Regarding the preparation of the urine sample and its microscopic examination, which of the following is *false*?
 - A. The sample should be examined within 3 hours of collection.
 - B. All crystals are birefringent under polarized light.
 - C. When collecting the urine sample, only the midstream urine should be collected.
 - D. Phase-contrast microscopy is helpful for particle identification.
3. The automated urine sediment analyzers available today:
 - A. are adequate for the evaluation of the kidney patient.
 - B. require high volumes of urine.
 - C. adequately identify renal tubular epithelial cells.
 - D. allow the examination of a high number of samples in a short period of time.

REFERENCES

- Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int Suppl.* 2013;3:1–150.
- Perazella MA. The urine sediment as a biomarker of kidney disease. *Am J Kidney Dis.* 2015;66:748–755.
- Becker GJ, Garigali G, Fogazzi GB. Advances in urine microscopy. *Am J Kidney Dis.* 2016;67:954–964.
- Tsai JJ, Yeun JY, Kumar VA, Don BR. Comparison and interpretation of urinalysis performed by a nephrologist versus a hospital-based clinical laboratory. *Am J Kidney Dis.* 2005;46:820–829.
- Clinical and Laboratory Standard Institute. *GP-16 A3. Urinalysis and Collection, Transportation, and Preservation of Urine Specimens: Approved Guideline.* 3rd ed. Wayne, PA: Clinical and Laboratory Standard Institute; 2009.
- Fogazzi GB. *The Urinary Sediment: An Integrated View.* 3rd ed. Milan: Elsevier; 2010.
- Lee W, Kim Y, Chang S, et al. The influence of vitamin C on the urine dipstick tests in the clinical specimens: a multicenter study. *J Clin Lab Anal.* 2017;32(5).
- Lam MO. False hematuria due to bacteriuria. *Arch Pathol.* 1995;119:717–721.
- Lamb EJ, MacKenzie F, Stevens PE. How should proteinuria be detected and measured? *Ann Clin Biochem.* 2009;46:205–217.
- Hermida FJ, Soto S, Benitez AJ. Evaluation of the urine/protein/creatinine ratio measured with the dipstick Clinitek Atlas PRO 12. *Clin Lab.* 2016;62:735–738.
- John KA, Cogswell ME, Campbell NR, et al. Accuracy and usefulness of select methods for assessing complete collection of 24-hour urine: a systematic review. *J Clin Hypertens.* 2016;18(5):456–467.
- Price CP, Newall R, Boyd JC. Use of protein/creatinine ratio measurements on random urine samples for prediction of significant proteinuria: a systematic review. *Clin Chem.* 2005;51:1577–1586.
- Hogan MC, Reich HN, Nelson PJ, et al. The relatively poor correlation between random and 24-hour urine protein excretion in patients with biopsy-proven glomerular diseases. *Kidney Int.* 2016;90(5):1080–1089.
- Shidham G, Ayoub I, Birmingham D, et al. Limited reliability of the spot urine protein/creatinine ratio in the longitudinal evaluation of patients with lupus nephritis. *Kidney Int Rep.* 2018;3(5):1057–1063.
- Matsushita K, Coresh J, Sang Y, et al. Estimated glomerular filtration rate and albuminuria for prediction of cardiovascular outcomes: a collaborative meta-analysis of individual participant data. *Lancet Diabetes Endocrinol.* 2015;3(7):514–525.
- Dimopoulos MA, Kastritis E, Terpos E, et al. International Myeloma Working Group recommendations for the diagnosis and management of myeloma-related renal impairment. *J Clin Oncol.* 2016;34(13):1544–1557.
- Beer JH, Vogt A, Neftel K, et al. False positive results for leucocytes in urine dipstick test with common antibiotics. *BMJ.* 1996;313:25.
- Ramlakhan SL, Burke DP, Goldman RS. Dipstick urinalysis for the emergency department evaluation of urinary tract infections in infants aged less than 2 years. *Eur J Emerg Med.* 2011;18:221–224.
- Fogazzi GB, Edefonti A, Garigali G, et al. Urine erythrocyte morphology in patients with microscopic haematuria caused by a glomerulopathy. *Pediatr Nephrol.* 2008;23:1093–1110.
- Fairley K, Birch DF. Hematuria: a simple method for identifying glomerular bleeding. *Kidney Int.* 1982;21:105–108.
- Rath B, Turner C, Hartley B, Chantler C. What makes red cells dysmorphic in glomerular hematuria? *Pediatr Nephrol.* 1992;6:424–427.
- Köhler H, Wandel E, Brunck B. Acanthocyturia: a characteristic marker for glomerular bleeding. *Kidney Int.* 1991;40:115–120.
- Tesser Poloni JA, Bosan IB, Garigali G, Fogazzi GB. Urinary red blood cells: not only glomerular and nonglomerular. *Nephron Clin Pract.* 2012;120:c36–c41.
- Fogazzi GB, Ferrari B, Garigali G, et al. Urinary sediment in acute interstitial nephritis. *Am J Kidney Dis.* 2012;60:330–332.
- Fogazzi GB, Saglimbeni L, Banfi G, et al. Urinary sediment features in proliferative and nonproliferative glomerular diseases. *J Nephrol.* 2005;18:703–710.
- Muriithi AK, Nasr SH, Leung N. Utility of urine eosinophils in the diagnosis of acute interstitial nephritis. *Clin J Am Soc Nephrol.* 2013;8:1857–1862.
- Hall IE, Coca SG, Perazella MA, et al. Risk of poor outcomes with novel and traditional biomarkers at clinical AKI diagnosis. *Clin J Am Soc Nephrol.* 2011;6:167–174.
- Fogazzi GB, Garigali G, Pieruzzi F, et al. A morphological classification of the fat particles found in the urinary sediment of patients with Fabry disease. *Clin Chem Lab Med.* 2021;59:1832–1843.
- Chawla LS, Domm A, Berger A, et al. Urinary sediment cast scoring index for acute kidney injury: a pilot study. *Nephron Clin Pract.* 2008;110:c145–c150.
- Spinelli D, Consonni D, Garigali G, Fogazzi GB. Waxy casts in the urinary sediment of patients with different types of glomerulonephritis: results of a prospective study. *Clin Chim Acta.* 2013;424:47–52.
- Luciano RL, Perazella MA. Crystalline-induced kidney disease: a case for urine microscopy. *Clin Kidney J.* 2015;8:131–136.
- Frucht V, Castiglione V, Lucas IT, et al. Advances in the identification of calcium carbonate urinary crystals. *Clin Chim Acta.* 2021;515:1–4.
- Baroni S, Garigali G, Primiano A, Gervasoni J, Fogazzi GB. Very unusual “needle-and pencil-like” uric acid crystals in the urine unmasked by infrared spectroscopy investigation. *Clin Chim Acta.* 2018;479:72–73.
- Covarelli C, Primiano A, Garigali G, et al. Other very rare and poorly known uric acid crystals for the first time also shown by polarized light and definitively identified by infrared spectroscopy. *Clin Chim Acta; Int J Clin Chem.* 2018;484:148–149.
- Wong KA, Pardy C, Pillay S, et al. Can the presence of crystalluria predict stone formation in patients with cystinuria? *J Endourol.* 2016;30:609–614.
- Garigali G, Marra G, Rizzo V, et al. A virtuous diagnostic and therapeutic roadmap triggered by a motivated and skilful urinary sediment examination. *Clin Chim Acta.* 2019;492:23–25.
- De Liso F, Garigali G, Ferraris Fusarini C, et al. How to identify sulfamethoxazole crystalluria. *Clin Chim Acta.* 2016;452:106–108.
- de Lastours V, Ferrari Rafael De Silva E, Daudon M, et al. High levels of atazanavir and darunavir in urine and crystalluria in asymptomatic patients. *J Antimicrob Chemother.* 2013;68:1850–1856.
- Daudon M, Frucht V, Bazin D, Jungers P. Drug-induced kidney stones and crystalline nephropathy: pathophysiology, prevention and treatment. *Drugs.* 2018;78(2):163–201.
- Daudon M, Hennequin C, Boujelben G, et al. Serial crystalluria determination and the risk of recurrence in calcium stone formers. *Kidney Int.* 2005;67:1934–1943.
- Kidney Disease: Improving Global Outcomes. KDIGO clinical practice guideline for the care of kidney transplant recipients. *Am J Transplant.* 2009;(suppl 31):S1–S157.
- Chakera A, Dyar OJ, Hughes E, et al. Detection of polyomavirus BK reactivation after renal transplantation using an intensive decoy cell surveillance program is cost-effective. *Transplantation.* 2011;93:1018–1023.
- Nickeleit V, True K, Detwiler T, et al. Risk assessment for polyomavirus nephropathy using urine cytology and the detection of decoy cells: cheap and efficient. *Transplantation.* 2012;94:e42–e44.
- Fogazzi GB, Cantù M, Saglimbeni L. Decoy cells” in the urine due to polyomavirus BK infection: easily seen by phase-contrast microscopy. *Nephrol Dial Transplant.* 2001;16:1496–1498.
- Singh HK, Bubendorf L, Mihatsch MJ, et al. Urine cytology of polyomavirus infection. *Adv Exp Med Biol.* 2006;577:201–212.
- Fogazzi GB, Pallotti F, Garigali G. Atypical/malignant urothelial cells in routine urinary sediment: worth knowing and reporting. *Clin Chim Acta.* 2015;439:107–111.

47. Previtali G, Ravasio R, Seghezzi M, Buoro S, Alessio MG. Performance evaluation of the new fully automated urine particle analyser UF-5000 compared to the reference method of the Fuchs-Rosenthal chamber. *Clin Chim Acta*. 2017;472:123–130.
48. İnce FD, Ellidağ HY, Koseoğlu M, et al. The comparison of automated urine analyzers with manual microscopic examination for urinalysis automated urine analyzers and manual urinalysis. *Practical laboratory medicine*. 2016;5:14–20.
49. Falbo R, Sala MR, Bussetti M, et al. Performance evaluation of a new and improved cuvette-based automated urinalysis analyzer with phase contrast microscopy. *Clin Chim Acta*. 2019;491:126–131.
50. Oyaert M, Speeckaert M, Boelens J, Delanghe JR. Renal tubular epithelial cells add value in the diagnosis of upper urinary tract pathology. *Clin Chem Lab Med (CCLM)*. 2020;58(4):597–604.